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CLAIM 1 -- Hydrostatic equilibrium of the plenum

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ODE : $w c^2 \frac{d\rho}{dr} = \rho \frac{GM}{r^2}$
Solution : $\text{Eq}(\rho(r), C1 \cdot \exp(-GM/(c^2 r w)))$
Boundary : $\rho \rightarrow \rho_0$ as $r \rightarrow \infty$: 1
[PASS] $\rho(r) = \rho_0 \exp[-GM/(w c^2 r)]$

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CLAIM 2 -- CRITICAL: exponential is NOT a 1/r power law

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$\rho(r)/\rho_0 = \exp(-A/r)$
Large-r expansion: $1 - A/r + A^2/(2r^2) + O(r^{-3}), (r, \infty)$
[NOTE] The 1/r dependence lies in the EXPONENT.
[NOTE] A 1/r *perturbation* emerges only at FIRST ORDER.
[NOTE] Correct phrasing: 'integrating the 1/r^2 force yields a
[NOTE] profile whose exponent depends on 1/r, which at first order
[NOTE] produces the required 1/r potential perturbation.'
[FAIL] if any manuscript says 'produces a 1/r profile' unqualified.

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CLAIM 3 -- Calibration of the equation of state, $w = 1/4$

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$n(r) = \sqrt{\rho_0/\rho} = \exp[GM/(2 w c^2 r)]$
First-order coefficient: $1/(2r w)$
Require $= 2/r$ (to match measured 1.75 arcsec) $\Rightarrow w = [1/4]$
[PASS] $w = 1/4$, i.e. $P = \rho c^2 / 4$
[NOTE] CALIBRATED, not derived. Epistemic status = Newton's G.
[NOTE] Open problem: why 1/4 and not 1/3 (isotropic photon gas)?

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CLAIM 4 -- Second-order divergence from GR (the falsifiable prediction)

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$n_{\text{ether}} = 2x^2 + 2x + 1$
 $n_{\text{GR}} = 7x^2/4 + 2x + 1$
2nd-order coefficients: ether = 2, GR = 7/4
[PASS] 1st order identical (both 2x) -- agrees with all existing tests
[PASS] 2nd order differs: 2 vs 7/4 (ratio 1.14286)
Solar limb $x = 2.1230e-06$; predicted difference = 0.232 microarcsec
[NOTE] Gaia precision ~20-25 uas. NOT currently testable.

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CLAIM 5 -- Light deflection, three routes

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Newtonian corpuscle : 0.8758 arcsec (historically refuted)
 Fluid/Fermat : 1.7516 arcsec
 GR null geodesic : 1.7516 arcsec (same closed form)
 Observed (VLBI) : 1.7500 +/- 0.0002 arcsec
 [PASS] fluid and GR agree exactly; within 1 sigma of observation
 [NOTE] NOT independent confirmation: w was calibrated on this datum.

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CLAIM 6 -- Mercury perihelion advance

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Delta phi = 6 pi GM/[a(1-e^2)c^2] per orbit
 Predicted : 42.992 arcsec/century (fluid Binet AND GR geodesic)
 Observed : 42.98 +/- 0.04 arcsec/century
 [PASS] within 1 sigma; genuinely independent of the w calibration

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CLAIM 7 -- Gravitational redshift and Shapiro delay

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Pound-Rebka $z = GM h / (c^2 R^2) = 2.4585e-15$
 Measured (Pound-Snider 1965) = 2.57e-15 +/- 0.26e-15
 [PASS] within 1 sigma

Shapiro round trip = $(4GM/c^3) \ln(4 r_1 r_2 / b^2) = 232.6 \text{ us}$
 Literature value ~240 us
 [PASS] consistent

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CLAIM 8 -- CRITICAL: frame dragging is TOPOLOGY match + FITTED prefactor

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Stokes rotating sphere : $v = \Omega R^3 \sin(\theta) / r^2$
 GR Lense-Thirring drag : $v = 2 G J \sin(\theta) / (c^2 r^2)$
 [PASS] identical DIPOLE TOPOLOGY: both go as $\sin(\theta)/r^2$
 [FAIL] claiming 'algebraically identical' physical fields.
 [NOTE] Stokes flow requires viscosity + no-slip boundary at $r=R$.
 [NOTE] Kerr is a VACUUM solution ($T_{\mu\nu} = 0$). Mechanisms differ.
 [NOTE] The prefactor is MATCHED by defining R_{eff} via
 [NOTE] $\Omega R_{\text{eff}}^3 = 2GJ/c^2$
 [NOTE] Computing the ratio after imposing this returns 1 BY
 [NOTE] CONSTRUCTION. That is calibration, not derivation.

Earth: $J = 5.8457e+33 \text{ kg m}^2/\text{s}$; $R_{\text{eff}} = 4919.6 \text{ m} = 4.92 \text{ km}$
 [NOTE] >>> R_{eff} IS A SECOND FITTED CONSTANT. <<<
 [NOTE] Any claim of 'a single calibrated constant' is FALSE.
 [NOTE] The series has TWO: w (static) and R_{eff} (rotational).

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CLAIM 9 -- PTA correlation functions (Paper on nanohertz background)

sep(deg)	Hellings-Downs	Scalar-transverse
0.01	+0.5000	+0.5000
10.00	+0.4425	+0.4981
30.00	+0.2116	+0.4833
49.20	+0.0011	+0.4567
60.00	-0.0824	+0.4375
82.60	-0.1519	+0.3911
90.00	-0.1449	+0.3750
121.80	+0.0001	+0.3091
150.00	+0.1697	+0.2667
180.00	+0.2500	+0.2500

HD minimum -0.1519 at 82.5 deg

HD zero crossings near 49.2 and 121.8 deg

ST range: 0.2500 to 0.5000 -- strictly positive

[PASS] HD is ANTI-correlated near 82.6 deg; ST never is.

[NOTE] This is a STRUCTURAL difference, not a coefficient mismatch.

[NOTE] Reported Bayes factor favouring HD over ST: ~2 to 2.5

[NOTE] Jeffreys scale 1-3 = 'barely worth mentioning'.

[NOTE] Plenum reading is DISFAVOURED but NOT excluded.

CLAIM 10 -- SMBHB orbital frequencies fall in the PTA band

M each	sep(pc)	period(yr)	f = 2/T (nHz)
1e+08	0.001	0.21	302.58
1e+08	0.010	6.62	9.57
1e+08	0.100	209.45	0.30
1e+09	0.001	0.07	956.85
1e+09	0.010	2.09	30.26
1e+09	0.100	66.23	0.96
1e+10	0.001	0.02	3025.84
1e+10	0.010	0.66	95.69
1e+10	0.100	20.94	3.03

[PASS] PTA band ~1-100 nHz is populated by SMBHB orbital timescales

[NOTE] True under EITHER interpretation; not discriminating.

DIMENSIONAL CONSISTENCY CHECKS

$\rho_{\text{ether}} = u/c^2$: $(\text{J}/\text{m}^3)/(\text{m}^2/\text{s}^2) = \text{kg}/\text{m}^3$ [PASS]

$L/(4 \pi r^2 c^3)$: $\text{W}/(\text{m}^2 * \text{m}^3/\text{s}^3) = \text{kg}/\text{m}^3$ [PASS]

$\text{GM}/(r c^2)$: dimensionless [PASS]

-> must MULTIPLY a density, never be ADDED to one.

$\omega_{\text{ether}} = (\rho c^5/\hbar)^{(1/4)}$: 1/s [PASS]

$u_{\text{CMB}} = a T^4 = 4.1717\text{e-}14 \text{ J/m}^3$
 $\rho_{\text{CMB}} = u/c^2 = 4.6417\text{e-}31 \text{ kg/m}^3$

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SUMMARY OF EPISTEMIC STATUS

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DERIVED : equilibrium profile; refractive index form;
 perihelion advance; redshift; Shapiro delay

CALIBRATED : $w = 1/4$ (from light deflection)

CALIBRATED : R_{eff} (from frame-dragging amplitude)

TOPOLOGY : frame-drag dipole shape ($\sin \theta / r^2$)

OPEN : w from first principles; Lorentz-invariance bounds;
 strong field; PTA polarization content

Two fitted constants. Not one. State this plainly.